

Simmon Yang

Toronto, ON

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SUMMARY

Electrical Engineering Technology student with hands-on experience in electrical systems, low-voltage infrastructure, renewable energy hardware, and field installation work. Strong background in troubleshooting, testing, and system integration, complemented by coursework in **PLC programming** and control systems. Seeking a Field Service / Controls role supporting complex automated equipment at customer sites.

TECHNICAL SKILLS

- **Electrical & Hardware:** DC/AC Circuits, Digital Circuits, Residential & Commercial Wiring, Low-Voltage Systems, Microcontrollers (Arduino, Raspberry Pi), PCB Design (KiCad), Security Systems, Network Cabling (Cat5e/Cat6/Cat6A), Soldering, Testing & Troubleshooting
- **Software & Tools:** Python, C, C++, Java, PHP, WordPress, MySQL, AutoCAD Electrical, Multisim, PLC Programming
- **Standards & Safety:** Electrical Code Comprehension, TIA/EIA Standards, WHMIS, Working at Heights, Lockout/Tagout

EDUCATION

Electrical Engineering Technology Diploma

Sep 2022 – Present

Centennial College

Scarborough, ON

- Relevant coursework: Electrical Theory, Monitoring Systems, Networking, **PLC Programming**, Power Generation, Microcontrollers, Signals & Communication
- GPA: 4.1

WORK EXPERIENCE

Electrician Assistant

Sep 2024 – Present

Independent / On-call

Greater Toronto Area, ON

- Assist licensed electricians with installation, maintenance, and repair of residential and commercial electrical systems, following drawings, codes, and site safety procedures including **lockout/tagout**.
- Pull, terminate, and label branch circuits and feeders for lighting, receptacles, and power distribution; install switches, outlets, lighting fixtures, and breakers, and verify operation using meters and functional tests.
- Support panel upgrades, rough-ins, grounding, and device installations, documenting work completed and communicating issues and status with supervisors and customers.

Low Voltage Technician

Jul 2025 – Sep 2025

OneNid Inc

Greater Toronto Area, ON

- Installed and terminated Cat5e/Cat6/Cat6A cabling for IP cameras, access points, and intercom systems; performed continuity and performance testing to ensure reliable system operation.
- Mounted and aligned IP and analog security cameras, configured DVR/NVR systems and network access, and troubleshoot wiring, device, and configuration faults on-site.
- Integrated access control hardware including card readers, maglocks, and motion sensors while adhering to TIA/EIA standards and customer requirements.

Electrical Technician Assistant

Jul 2021 – Aug 2021

Microgreen Solar Corp

Markham, ON

- Repaired and tested DC-AC inverter power packs for renewable energy systems, using meters and load testing to verify performance and resolve faults.
- Performed component-level troubleshooting and soldering on PCBs, following documented procedures and maintaining detailed test records.

PROJECTS

Automated Circadian Lighting System (Retrofit)

Sep 2025 – Dec 2025

Lighting Control / Energy Efficiency

- Designed a retrofit automated LED lighting system to adjust intensity and color temperature over the day, improving visual comfort and energy efficiency while integrating with existing fixtures.
- Developed control logic and performed measurements to demonstrate stable operation and reduced energy usage, applying principles similar to field commissioning and system tuning.

Adaptive LED Headlight Simulation

Sep 2025 – Dec 2025

PIC16F747 / Embedded Systems

- Implemented a microcontroller-based adaptive headlight simulation using a PIC16F747 and ultrasonic sensing to adjust LED patterns based on distance.
- Programmed embedded **C** control logic for real-time sensing, decision-making, and output control, validating concepts of automation, safety, and diagnostics.

8x8x8 RGB LED Cube

Dec 2021 – Jun 2023

Arduino / PCB Design

- Designed and built a 3D RGB LED cube controlled by an Arduino microcontroller, including custom PCB design, assembly, and debugging of multiplexed LED control circuits.

CERTIFICATIONS

- WHMIS Certification (2024)
- Working at Heights Certification (2024)
- G3 Gas Technician (2025)